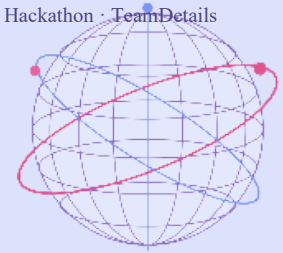
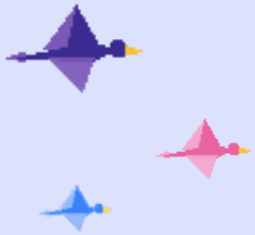
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · TeamDetails





Participant Details

PROJECT TITLE
CARBONWISE - SmartEnergyOptimization

TRACK - 06
Quantum Applications for Climate Change & Sustainability

TEAM NAME
Q-Rious Minds

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INSTITUTION / ORGANISATION
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COUNTRY
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Problem Statement

"University campuses face challenges in efficiently managing solar energy, battery storage, and grid electricity due to fluctuating solar generation and changing electricity demand. This can increase electricity costs and carbon emissions."

Objectives & Novelty

Objectives

- AddOptimizebatterychargingand discharging schedules.
- Maximize solar energy utilization and reduce grid dependency.
- Minimize electricity costs and estimated carbon emissions.
- Implement QAOA using Qiskit for energy optimization.
- Compare quantum and classical optimization results.
- Visualize results through an interactive dashboard.

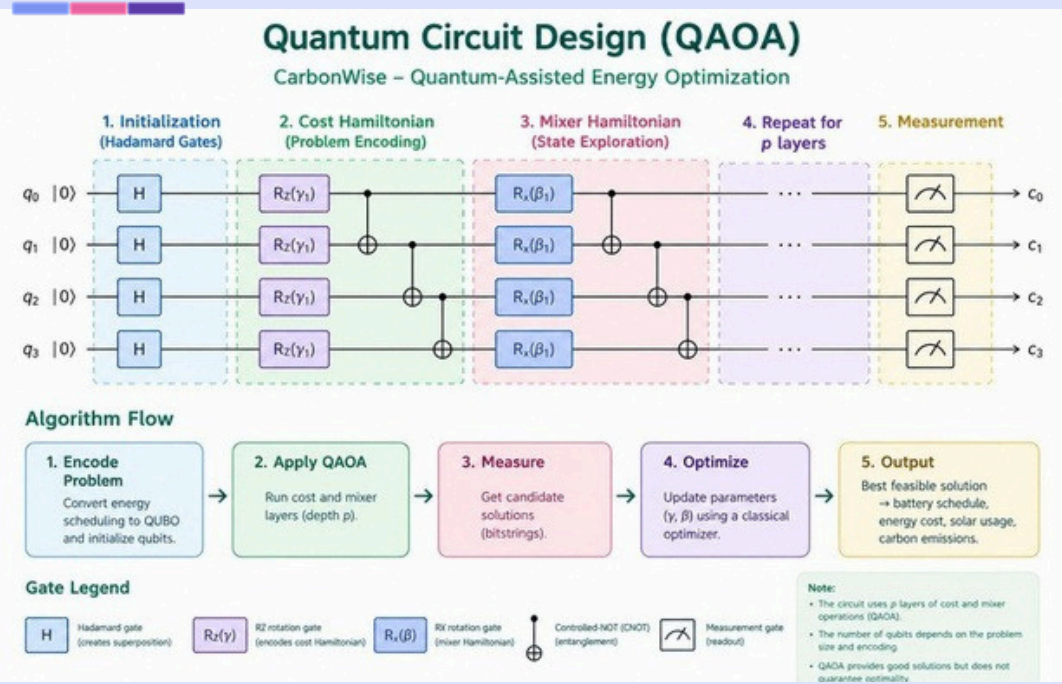
Novelty

- Quantum-Assisted Energy Management: Applies QAOA to campus energy scheduling.
- Hybrid Comparison: Evaluates quantum and classical optimization methods on the same problem.
- Multi-Objective Optimization: Considers electricity costs, solar utilization, and estimated carbon emissions.
- Interactive Visualization: Presents energy schedules and performance comparisons through a unified dashboard.

Solution & Quantum Advantage

CarbonWise is a quantum-assisted energy optimization system that uses Qiskit's Quantum Approximate Optimization Algorithm (QAOA) to optimize battery charging schedules, maximize solar energy utilization, and reduce grid electricity costs and estimated carbon emissions. It compares quantum and classical optimization results and presents them through an interactive dashboard and a generated demonstration video.

Algorithm & Circuit Design



- *Input: Solar generation, campus energy demand, electricity prices, and battery constraints.*
- Formulation: Convert the energyscheduling problem into a Quadratic Unconstrained Binary Optimization (QUBO) model.
- QAOA Implementation: Use Qiskit to construct and run the Quantum Approximate Optimization Algorithm.
- Classical Optimization: Solve the same problem using a classical optimizer for comparison.
- Evaluation: Compare energy cost, solar utilization, grid consumption, and estimated carbon emissions.
- Output: Generate an optimized battery schedule and display results on the dashboard.

Implementation & Tech Stack

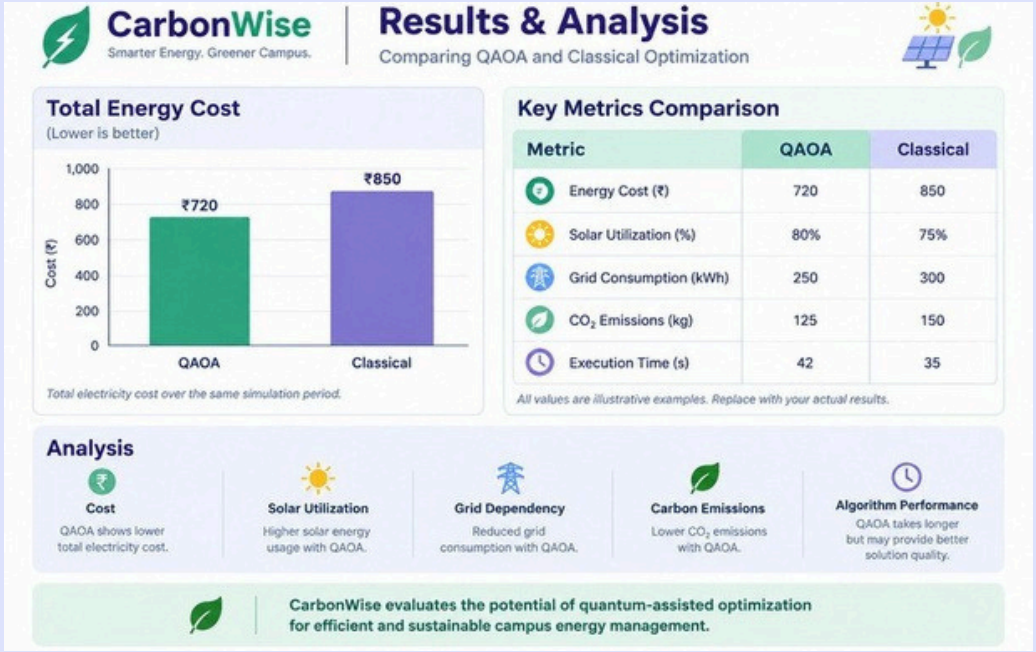
IMPLEMENTATION

Economic:Potentially reduces campus electricitycosts.
Environmental: Helps lower estimated carbon emissions.
Energy Efficiency: Improves solar utilization and battery scheduling.
Technological: Demonstrates quantum-assisted energy optimization using QAOA.
Sustainability: Supports greener and smarter university campuses.

TECH STACK

Python: Coreprogrammingand optimization logic.
Qiskit: Quantum circuit design and QAOA implementation.
Classical Optimizer: Baseline comparison with QAOA.
Pandas & NumPy: Energy data processing and calculations.
Google Colab / Jupyter Notebook: Development and testing environment.

Results & Analysis



Results:
Optimized battery charging and discharging schedules.
Evaluated solar energy utilization and grid consumption.
Compared electricity costs using QAOA and classical optimization.

Analysis:
Cost: Compare total electricity costs for both methods.
Solar Utilization: Measure the effective use of solar power.

Impact & Real-World Applications

IMPACT

Economic Impact:Potentialreduction in campus electricity costs.
Environmental Impact: Lower estimated carbon emissions through improved energy management.
Energy Efficiency: Better solar energy utilization and battery scheduling
Technological Impact: Demonstrates the application of QAOA and Qiskit to energy optimization.

CarbonWise – Real-World Applications

Universities and Colleges: Optimizesolar power,batterystorage, and campus electricity usage
.Smart Buildings: Manage energy demand in offices, libraries, and laboratories.
Microgrids: Coordinate renewable energy, battery storage, and grid electricity
.Solar-Powered Campuses: Maximize solar energy utilization and reduce grid dependency.
Commercial Facilities: Reduce electricity expenses

Challenges & Future Scope

CHALLENGES

Data Accuracy:Solar generation andcampus energy demand fluctuateover time

.Quantum Limitations: QAOA performance depends on circuit depth, qubit availability, and optimization parameters.

Complex Optimization: Battery constraints and energy scheduling increase problem complexity.

Classical Benchmarking: Ensuring a fair comparison between quantum and classical methods can be challenging.

FUTURE SCOPE

Real-Time Integration: Connect smart meters,solar panels, and battery management systems.AI-Based Forecasting: Predict solar generation and campus energy demand using machine learning.

Quantum Hardware: Test QAOA on real quantum processors as hardware capabilities improve.

Scalable Deployment: Extend CarbonWise to smart buildings, university campuses, and microgrids.

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.

GITHUB REPOSITORY
2https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-q-rious-minds

DEMO VIDEO LINK
https://youtu.be/dN97MQCEJTY

